

Lecture 1

Introduction to Dynamics

Prof. Pilwon Hur
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1 Introduction

Introduction

In this course, we will study dynamics. In general, dynamics is a study of moving objects.
That is, movement of objects, the cause of the movement will be covered.



1 Introduction

What is Dynamics?

⚙️ Dynamics

- Dynamics is one of the most important subjects in Physics that studies the movement of objects under the influence of forces.



1 Introduction

What is Dynamics?

⚙️ Dynamics

- Mechanics is a study of how objects deform under the influence of forces. Specifically, solid mechanics examines how solid materials deform when forces are applied. Fluid mechanics investigates how fluids (e.g., water, air) move and change its shape under forces.



1 Introduction

What is Dynamics?

⚙️ Dynamics

- Therefore, solid mechanics and fluid mechanics can also be part of dynamics.



1 Introduction

Scope of this Course

⚙️ Dynamics

- Even though Dynamics includes studies of any moving objects under forces, solid mechanics and fluid mechanics will be excluded from this course:

Solid mechanics

Deformation of solid materials is small, and the effect of deformation on the movement is minimal.

Fluid mechanics

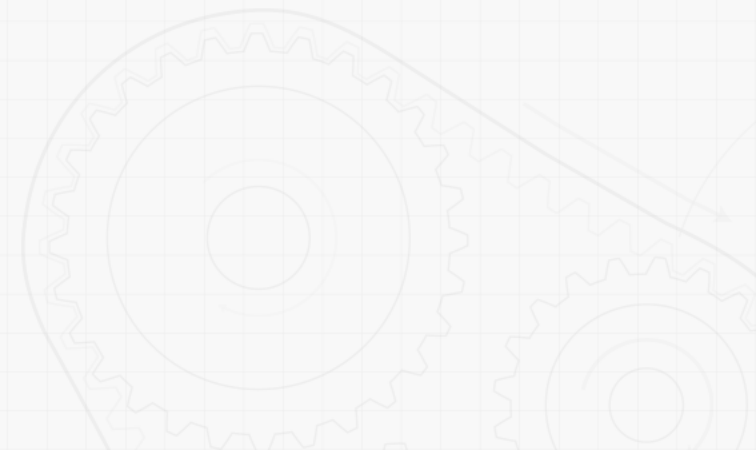
Fluid deforms significantly when forces are applied to it. However, there are significant relative movement between particles (e.g., water molecules).

1 Introduction

Scope of this Course

⚙ Dynamics

- Since the relative movement itself and the related interacting forces are quite complicated, different approaches are used compared to rigid body dynamics.



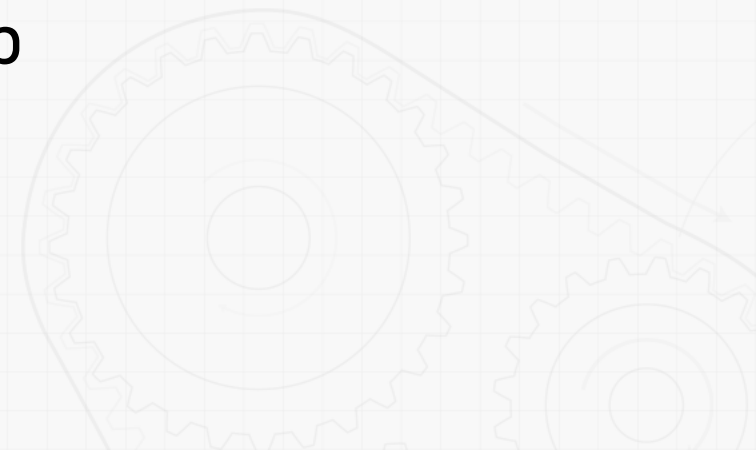
1 Introduction

Scope of this Course

⚙️ Particles or Rigid Bodies

- In this course (i.e., entry level dynamics course), we will limit the scope of materials to **particles or rigid bodies**. Rigid bodies are ideal materials that do not deform at all.

Also, particles are assumed to have masses only, but no volume at all. These assumptions will make it much easier for undergraduate students to grasp the important concepts of dynamics.



Some of the Principles used in Dynamics

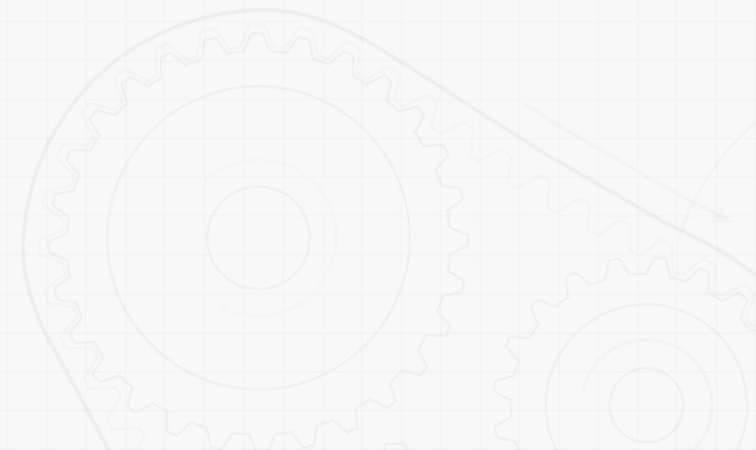
⚙ Principles

**Newton's law of
motion
(2nd)**

**Newton's law of
action/reaction
(3rd)**

**Coulomb's
friction law
(contact vs. force)**

**Hooke's law of
deformation
(deformation vs. force)**



Some of the Principles used in Dynamics

⚙ Principles

**Viscous
damping**
(speed vs. force)

Electromagnetics

Piezoelectricity
(deformation vs.
electricity)

Seebeck effect
(temperature vs.
electricity)

...

Some of Mathematics Tools used in Dynamics

⚙ Mathematics Tools

Vectors

Integration

**Scalar
Differentiation**

**Vector
Differentiation**

1 Introduction

Key Topics in Dynamics

⚙️ Key Topics

Kinematics

How objects move



Keywords:

- position
- velocity
- acceleration
- angular velocity
- angular acceleration

Kinetics

What causes the movement of objects



Keywords:

- forces
- torques

1 Introduction

Key Topics in Dynamics

⚙️ Key Topics

- Other important terminologies

Energy

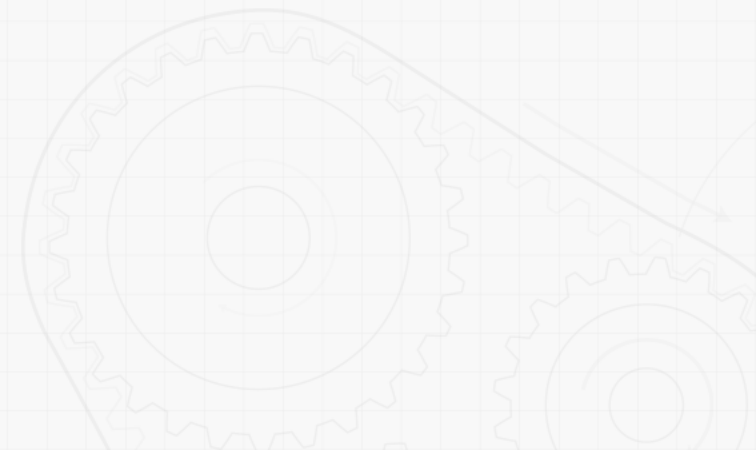
Potential energy,
kinetic energy

Vector-valued

momentum,
angular momentum

Geometry-related

moment of inertia,
reference frame,
coordinate systems,
center of mass



1 Introduction

Minimal Mathematics Review

⚙️ Key Topics

- Differentiation

https://en.wikipedia.org/wiki/Differentiation_rules

- Integration

<https://en.wikipedia.org/wiki/Integral>

- Vector

https://en.wikipedia.org/wiki/Vector_space

- Mathematica tutorial

<https://www.wolfram.com/language/fast-introduction-for-math-students/en/>

References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, McGraw Hill
- 유홍희, "동역학", 1판, 문운당